

Лабораторная работа

Номер 13

Андрюшин Н. С.

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Российский университет дружбы народов, Москва, Россия

Информация

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Приобретение навыков настройки сервера NFS для удалённого доступа к ресурсам

Установка пакета nfs-utils на сервере

```
[nsandryushin@server nsandryushin.net ~]$ sudo dnf -y install nfs-utils
[sudo] password for nsandryushin:
Extra Packages for Enterprise Linux 10 - x86_64                21 kB/s | 39 kB    00:01
Extra Packages for Enterprise Linux 10 - x86_64                1.6 MB/s | 5.5 MB    00:03
Rocky Linux 10 - BaseOS                                         385 B/s | 4.3 kB    00:14
Rocky Linux 10 - BaseOS                                         285 kB/s | 1.8 MB    00:06
Rocky Linux 10 - AppStream                                       8.3 kB/s | 4.3 kB    00:00
Rocky Linux 10 - AppStream                                       960 kB/s | 1.9 MB    00:02
Rocky Linux 10 - CRB                                             2.4 kB/s | 4.3 kB    00:01
Rocky Linux 10 - CRB                                             238 kB/s | 480 kB    00:02
Rocky Linux 10 - Extras                                          1.2 kB/s | 3.1 kB    00:02
Rocky Linux 10 - Extras                                          1.4 kB/s | 4.7 kB    00:03
Dependencies resolved.
=====
Package                Architecture      Version           Repository         Size
=====
Installing:
nfs-utils               x86_64            1:2.8.3-0.el10    baseos             475 k
Upgrading:
libipa_hbac             x86_64            2.11.1-2.el10.1.1 baseos             34 k
libldb                  x86_64            4.22.4-106.el10   baseos            181 k
libnss_smbclient        x86_64            4.22.4-106.el10   baseos             75 k
libnss_certmap         x86_64            2.11.1-2.el10.1.1 baseos             81 k
libnss_idmap            x86_64            2.11.1-2.el10.1.1 baseos             41 k
libnss_nss_idmap        x86_64            2.11.1-2.el10.1.1 baseos             44 k
libnss_sudo             x86_64            2.11.1-2.el10.1.1 baseos             33 k
libtalloc               x86_64            2.4.3-100.el10    baseos             33 k
libtdb                  x86_64            1.4.13-100.el10   baseos             55 k
libtevent               x86_64            0.16.2-100.el10   baseos             50 k
libwbclient             x86_64            4.22.4-106.el10   baseos             43 k
samba-client-libs       x86_64            4.22.4-106.el10   baseos             5.3 M
samba-common            noarch            4.22.4-106.el10   baseos            174 k
samba-common-libs       x86_64            4.22.4-106.el10   baseos            104 k
sssd                    x86_64            2.11.1-2.el10.1.1 baseos             25 k
sssd-ad                 x86_64            2.11.1-2.el10.1.1 baseos            195 k
sssd-client             x86_64            2.11.1-2.el10.1.1 baseos            152 k
sssd-common             x86_64            2.11.1-2.el10.1.1 baseos             1.5 M
sssd-common-pac         x86_64            2.11.1-2.el10.1.1 baseos             89 k
sssd-ipa                x86_64            2.11.1-2.el10.1.1 baseos            274 k
sssd-krb                x86_64            2.11.1-2.el10.1.1 baseos            103 k
sssd-krb5               x86_64            2.11.1-2.el10.1.1 baseos             62 k
sssd-krb5-common        x86_64            2.11.1-2.el10.1.1 baseos             93 k
sssd-ldap               x86_64            2.11.1-2.el10.1.1 baseos            132 k
sssd-proxy              x86_64            2.11.1-2.el10.1.1 baseos             70 k
```

Рис. 1: Установка пакета nfs-utils на сервере

Создание каталога и открытие файла конфигурации

```
[nsandryushin@server.nsandryushin.net ~]$ sudo mkdir -p /srv/nfs  
[nsandryushin@server.nsandryushin.net ~]$ sudo nano /etc/exports
```

Рис. 2: Создание каталога и открытие файла конфигурации

Настройка экспорта каталога



```
GNU nano 8.1 /etc/exports
/srv/nfs *(ro)
```

Рис. 3: Настройка экспорта каталога

Настройка SELinux, запуск служб и настройка firewall

```
[nsandryushin@server.nsandryushin.net ~]$ sudo semanage fcontext -a -t nfs_t "/srv/nfs(/.*)?"
[nsandryushin@server.nsandryushin.net ~]$ sudo restorecon -vR /srv/nfs
Relabeled /srv/nfs from unconfined_u:object_r:var_t:s0 to unconfined_u:object_r:nfs_t:s0
[nsandryushin@server.nsandryushin.net ~]$ systemctl start nfs-server.service
[nsandryushin@server.nsandryushin.net ~]$ systemctl enable nfs-server.service
Created symlink '/etc/systemd/system/multi-user.target.wants/nfs-server.service' → '/usr/lib/systemd/system/nfs-server.service'.
[nsandryushin@server.nsandryushin.net ~]$ sudo firewall-cmd --add-service=nfs
success
[nsandryushin@server.nsandryushin.net ~]$ sudo firewall-cmd --add-service=nfs --permanent
success
[nsandryushin@server.nsandryushin.net ~]$ sudo firewall-cmd --reload
success
[nsandryushin@server.nsandryushin.net ~]$
```

Рис. 4: Настройка SELinux, запуск служб и настройка firewall

Установка nfs-utils на клиенте

```
[nsandyushin@client nsandyushin.net ~]$ sudo dnf -y install nfs-utils
[sudo] password for nsandyushin:
Extra Packages for Enterprise Linux 10 - x86_64          9.0 kB/s | 39 kB    00:04
Extra Packages for Enterprise Linux 10 - x86_64          2.2 MB/s | 5.5 MB    00:02
Rocky Linux 10 - BaseOS                                   2.4 kB/s | 4.3 kB    00:01
Rocky Linux 10 - BaseOS                                   319 kB/s | 1.0 MB    00:05
Rocky Linux 10 - AppStream                                821 B/s | 4.3 kB    00:05
Rocky Linux 10 - AppStream                               179 kB/s | 1.9 MB    00:11
Rocky Linux 10 - CRB                                       7.5 kB/s | 4.3 kB    00:00
Rocky Linux 10 - CRB                                       12 kB/s | 400 kB    00:40
Rocky Linux 10 - Extras                                    1.2 kB/s | 3.1 kB    00:02
Rocky Linux 10 - Extras                                    2.4 kB/s | 4.7 kB    00:01
Dependencies resolved.
=====
Package              Architecture      Version              Repository           Size
=====
Installing:
nfs-utils            x86_64            1:2.8.3-0.el10      baseos               475 k
Upgrading:
libipa_hbac          x86_64            2.11.1-2.el10.1.1   baseos               34 k
libldb               x86_64            4.22.4-106.el10     baseos               181 k
libnss_smbclient     x86_64            4.22.4-106.el10     baseos               75 k
libsss_certmap       x86_64            2.11.1-2.el10.1.1   baseos               61 k
libsss_idmap         x86_64            2.11.1-2.el10.1.1   baseos               41 k
libsss_nss_idmap     x86_64            2.11.1-2.el10.1.1   baseos               44 k
libsss_sudo          x86_64            2.11.1-2.el10.1.1   baseos               33 k
libtalloc            x86_64            2.4.3-100.el10      baseos               33 k
libtdb               x86_64            1.4.13-100.el10     baseos               55 k
libtevent            x86_64            0.16.2-100.el10     baseos               50 k
libwbclient          x86_64            4.22.4-106.el10     baseos               43 k
samba-client-libs    x86_64            4.22.4-106.el10     baseos               5.3 M
samba-common         noarch            4.22.4-106.el10     baseos               174 k
samba-common-libs    x86_64            4.22.4-106.el10     baseos               104 k
sssd                 x86_64            2.11.1-2.el10.1.1   baseos               25 k
sssd-ad              x86_64            2.11.1-2.el10.1.1   baseos               195 k
sssd-client          x86_64            2.11.1-2.el10.1.1   baseos               152 k
sssd-common          x86_64            2.11.1-2.el10.1.1   baseos               1.5 M
sssd-common-pac      x86_64            2.11.1-2.el10.1.1   baseos               89 k
sssd-ipa             x86_64            2.11.1-2.el10.1.1   baseos               274 k
sssd-krb5            x86_64            2.11.1-2.el10.1.1   baseos               103 k
sssd-krb5-common     x86_64            2.11.1-2.el10.1.1   baseos               62 k
sssd-krb5-common     x86_64            2.11.1-2.el10.1.1   baseos               93 k
sssd-ldap            x86_64            2.11.1-2.el10.1.1   baseos               132 k
sssd-proxy           x86_64            2.11.1-2.el10.1.1   baseos               70 k
```

Рис. 5: Установка nfs-utils на клиенте

Ошибка подключения RPC при просмотре ресурсов на клиенте

```
[nsandryushin@client.nsandryushin.net ~]$ sudo showmount -e server.nsandryushin.net  
clnt_create: RPC: Unable to receive  
[nsandryushin@client.nsandryushin.net ~]$ █
```

Рис. 6: Ошибка подключения RPC при просмотре ресурсов на клиенте

Остановка межсетевого экрана на сервере

```
[nsandryushin@server.nsandryushin.net ~]$ systemctl stop firewalld.service  
[nsandryushin@server.nsandryushin.net ~]$ █
```

Рис. 7: Остановка межсетевого экрана на сервере

Успешный просмотр списка экспорта при отключенном фаерволе

```
[nsandryushin@client.nsandryushin.net ~]$ sudo showmount -e server.nsandryushin.net
Export list for server.nsandryushin.net:
/srv/nfs *
[nsandryushin@client.nsandryushin.net ~]$
```

Рис. 8: Успешный просмотр списка экспорта при отключенном фаерволе

Запуск firewalld и просмотр открытых портов

```
[nsandryushin@server.nsandryushin.net ~]$ systemctl start firewalld
[nsandryushin@server.nsandryushin.net ~]$ lsof | grep TCP
firefox      8766                nsandryushin  143u      IPv4           35790      0t0      TCP serv
er.nsandryushin.net:46512->34.107.243.93:https (ESTABLISHED)
firefox      8766  8786 firefox      nsandryushin  143u      IPv4           35790      0t0      TCP serv
er.nsandryushin.net:46512->34.107.243.93:https (ESTABLISHED)
firefox      8766  8787 WaylandPr nsandryushin  143u      IPv4           35790      0t0      TCP serv
er.nsandryushin.net:46512->34.107.243.93:https (ESTABLISHED)
```

Рис. 9: Запуск firewalld и просмотр открытых портов

Службы rpcbind и mountd в выводе lsof

```
rpcbind 10983          rpc      5u      IPv4      44875      0t0      TCP   *:s
unrpc (LISTEN)
rpcbind 10983          rpc      7u      IPv6      47113      0t0      TCP   *:s
unrpc (LISTEN)
rpc.statd 10984        rpcuser  9u      IPv4      52666      0t0      TCP   *:3
8465 (LISTEN)
rpc.statd 10984        rpcuser 11u      IPv6      52673      0t0      TCP   *:5
4055 (LISTEN)
rpc.mount 11013        root     5u      IPv4      53817      0t0      TCP   *:m
ountd (LISTEN)
rpc.mount 11013        root     7u      IPv6      53830      0t0      TCP   *:m
ountd (LISTEN)
[nsandryushin@server.nsandryushin.net ~]$
```

Рис. 10: Службы rpcbind и mountd в выводе lsof

Просмотр UDP-соединений с помощью lsof (начало вывода)

```
[nsandryushin@server.nsandryushin.net ~]$ sudo lsof | grep UDP
lsof: WARNING: can't stat() fuse.gvfsd-fuse file system /run/user/1001/gvfs
Output information may be incomplete.
lsof: WARNING: can't stat() fuse.portal file system /run/user/1001/doc
Output information may be incomplete.
systemd      1                root    384u      IPv4        47104      0t0      UDP *:s
unrpc
systemd      1                root    400u      IPv6        47122      0t0      UDP *:s
unrpc
avahi-daemon 884             avahi    12u      IPv4        9422       0t0      UDP *:m
dns
avahi-daemon 884             avahi    13u      IPv6        9423       0t0      UDP *:m
dns
chronyd       941             chrony    5u      IPv4        9447       0t0      UDP loc
alhost:323
chronyd       941             chrony    6u      IPv6        9448       0t0      UDP loc
alhost:323
chronyd       941             chrony    7u      IPv4        9449       0t0      UDP *:n
tp
named        1305            named     6u      IPv4        25788      0t0      UDP ns.
nsandryushin.net:domain
named        1305            named    25u      IPv4        10146      0t0      UDP loc
alhost:domain
```

Рис. 11: Просмотр UDP-соединений с помощью lsof (начало вывода)

Просмотр UDP-соединений с помощью lsof (окончание вывода)

```
rpcbind 10983          rpc      6u      IPv4      47104      0t0      UDP *:s
unrpc
rpcbind 10983          rpc      8u      IPv6      47122      0t0      UDP *:s
unrpc
rpc.statd 10984        rpcuser  5u      IPv4      53653      0t0      UDP loc
alhost:984
rpc.statd 10984        rpcuser  8u      IPv4      53674      0t0      UDP *:3
8992
rpc.statd 10984        rpcuser 10u      IPv6      52670      0t0      UDP *:4
1499
rpc.mount 11013        root     4u      IPv4      53814      0t0      UDP *:m
ountd
rpc.mount 11013        root     6u      IPv6      53827      0t0      UDP *:m
ountd
[nsandryushin@server.nsandryushin.net ~]$
```

Рис. 12: Просмотр UDP-соединений с помощью lsof (окончание вывода)

Настройка firewalld для работы служб mountd и rpc-bind

```
[nsandryushin@server nsandryushin.net ~]$ firewall-cmd --get-services
@-AD RH-Satellite-6 RH-Satellite-6-capsule afp alvr amanda-client amanda-k5-client anqp amqps anno-1602 a
nno-1800 apcupsd aseqnet audit ausweisapp2 bacula bacula-client bareos-director bareos-filedemon bareos-
storage bb bgp bitcoin bitcoin-rpc bitcoin-testnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-exporter
ceph-mon cfengine checkmk-agent civilization-iv civilization-v cockpit collectd condor-collector cratedb
ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-quic dns-over-tls docker
-registry docker-swarm dropbox-lansync elasticsearch etcd-client etcd-server factorio finger foreman fore
man-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera ganglia-clie
n t ganglia-master git gpsd grafana gre high-availability http http3 https ident imap imaps iperf2 iperf3 i
pps ippp ippp-client ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kibana klogon kpas
swd kprop kshell kube-api kube-apiserver kube-control-plane kube-control-plane-secure kube-controller-man
ager kube-controller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler-secure kube-work
er kubelet kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-network llmr llmr-c
lient llmr-tcp llmr-udp managesieve matrix ndns memcache minecraft minddlna mdp mongodb mosh mountd mp
d mqtt mqtt-tls ms-wbt mssql murmur mysql nbd nebula need-for-speed-most-wanted netbios-ns netdata-dashbo
ard nfs nfs3 nmap-0183 nrpe ntp nut opentelemetry openvpn ovirt-imageio ovirt-storageconsole ovirt-vncon
sole plex pncd pnpnproxy pwebapi pwebapis pop3 pop3s postgresql privoxy prometheus prometheus-node-exporte
r proxy-dhcp ps2link ps3netrvr ptp pulseaudio puppetmaster quassel radius radsec rdp redis redis-sentinel
rootd rpc-bind rquoted rsh rsyncd rtsp salt-master samba samba-client samba-dc sane settlers-history-col
lection slp slimevr slp smtp smtp-submission smtps snmp snmptls snmptls-trap snmptrap spiderOak-lans
ync spotify-sync sound sddp ssh ssh-cusom statshr steam-lan-transfer steam-streaming stellaris stronghol
d-crusader stun stuns submission supertuxkart svdrp svn syncthing syncthing-gui syncthing-relay synergy s
yscomlan syslog syslog-tls telnet tentacle terraria tftp tile38 tinc tor-socks transmission-client turn t
urns upnp-client vdsu vnc-server vrrp warpinator wben-http wben-https wireguard ws-discovery ws-discovery
-client ws-discovery-host ws-discovery-tcp ws-discovery-udp wssd wssd-http wsmn wsmans xdncc xmp-bosh x
mp-client xmp-local xmp-server zabbix-agent zabbix-java-gateway zabbix-server zabbix-trapper zabbix-we
b-service zero-k zerotier
[nsandryushin@server nsandryushin.net ~]$ sudo firewall-cmd --add-service=mountd --add-service=rpc-bind
success
[nsandryushin@server nsandryushin.net ~]$ sudo firewall-cmd --add-service=mountd --add-service=rpc-bind --
-permanent
success
[nsandryushin@server nsandryushin.net ~]$ sudo firewall-cmd --reload
success
[nsandryushin@server nsandryushin.net ~]$
```

Рис. 13: Настройка firewalld для работы служб mountd и rpc-bind

Проверка экспорта NFS-ресурса командой showmount

```
[nsandryushin@client.nsandryushin.net ~]$ sudo showmount -e server.nsandryushin.net
[sudo] password for nsandryushin:
Export list for server.nsandryushin.net:
/srv/nfs *
[nsandryushin@client.nsandryushin.net ~]$
```

Рис. 14: Проверка экспорта NFS-ресурса командой showmount

Монтирование каталога NFS на клиенте и проверка через mount

```
[nsandryushin@client nsandryushin.net ~]$ sudo mkdir -p /mnt/nfs
[nsandryushin@client nsandryushin.net ~]$ sudo mount server.nsandryushin.net:/srv/nfs /mnt/nfs
[nsandryushin@client nsandryushin.net ~]$ mount
/dev/mapper/cl_vbox-root on / type xfs (rw,relatime,seclabel,attr2,inode64,logbufs=8,logbsize=32k,noquota)
)
devtmpfs on /dev type devtmpfs (rw,nosuid,seclabel,size=4896k,nr_inodes=211266,mode=755,inode64)
tmpfs on /dev/shm type tmpfs (rw,nosuid,nodev,seclabel,inode64)
devpts on /dev/pts type devpts (rw,nosuid,noexec,relatime,seclabel,gid=5,mode=620,ptmxmode=000)
sysfs on /sys type sysfs (rw,nosuid,nodev,noexec,relatime,seclabel)
securityfs on /sys/kernel/security type securityfs (rw,nosuid,nodev,noexec,relatime)
cgroupl on /sys/fs/cgroup type cgroup2 (rw,nosuid,nodev,noexec,relatime,seclabel,nodeset,memory_recursive)
vproc on /sys/fs/vproc type vproc (rw,nosuid,nodev,noexec,relatime,seclabel)
efivarfs on /sys/firmware/efi/efivars type efivarfs (rw,nosuid,nodev,noexec,relatime)
bpf on /sys/fs/bpf type bpf (rw,nosuid,nodev,noexec,relatime,mode=780)
configfs on /sys/kernel/config type configfs (rw,nosuid,nodev,noexec,relatime)
proc on /proc type proc (rw,nosuid,nodev,noexec,relatime)
tmpfs on /run type tmpfs (rw,nosuid,nodev,seclabel,size=345560k,nr_inodes=819200,mode=755,inode64)
selinuxfs on /sys/fs/selinux type selinuxfs (rw,nosuid,noexec,relatime)
systemd on /proc/sys/fs/binfmt_misc type binfmt_misc (rw,relatime,fd=36,pgrp=1,timeout=0,skipno=5,maxproto=5,direct,pipe_ino=5536)
mqueue on /dev/mqueue type mqueue (rw,nosuid,nodev,noexec,relatime,seclabel)
debugfs on /sys/kernel/debug type debugfs (rw,nosuid,nodev,noexec,relatime,seclabel)
hugetlbfs on /dev/hugepages type hugetlbfs (rw,nosuid,nodev,relatime,seclabel,pagesize=2M)
tracfs on /sys/kernel/tracing type tracfs (rw,nosuid,nodev,noexec,relatime,seclabel)
tmpfs on /run/credentials/systemd-journald.service type tmpfs (rw,nosuid,nodev,noexec,relatime,noexecfo)
w,seclabel,size=1024k,nr_inodes=1024,mode=780,inode64,noswap)
fusectl on /sys/fs/fuse/connections type fusectl (rw,nosuid,nodev,noexec,relatime)
/dev/sda2 on /boot type xfs (rw,relatime,seclabel,attr2,inode64,logbufs=8,logbsize=32k,noquota)
/dev/sda1 on /boot/efi type vfat (rw,relatime,fmask=0077,dmask=0077,codepage=437,iocharset=ascii,shortnam)
e=wint.errors=remount-ro)
/dev/mapper/cl_vbox-home on /home type xfs (rw,relatime,seclabel,attr2,inode64,logbufs=8,logbsize=32k,no)
quota)
vagrant on /vagrant type vboxsf (rw,nosuid,relatime,iocharset=utf8,uid=1000,gid=1000)
tmpfs on /run/user/1001 type tmpfs (rw,nosuid,nodev,relatime,seclabel,size=172776k,nr_inodes=43194,mode=7)
00,uid=1001,gid=1001,inode64)
gvfsd-fuse on /run/user/1001/gvfs type fuse.gvfsd-fuse (rw,nosuid,nodev,relatime,user_id=1001,group_id=10)
01)
vagrant on /vagrant type vboxsf (rw,nosuid,relatime,iocharset=utf8,uid=1000,gid=1000,netdev)
portal on /run/user/1001/doc type fuse.portal (rw,nosuid,nodev,relatime,user_id=1001,group_id=1001)
tmpfs on /run/user/0 type tmpfs (rw,nosuid,nodev,relatime,seclabel,size=172776k,nr_inodes=43194,mode=700,
inode64)
server.nsandryushin.net:/srv/nfs on /mnt/nfs type nfs4 (rw,relatime,vers=4.2,rsize=262144,wsize=262144,na)
mlen=755,hard,proto=tcp,timeo=600,retrans=2,sec=sys,clientaddr=192.168.1.30,local_lock=none,addr=192.168.
1.3)
[nsandryushin@client nsandryushin.net ~]$
```

Рис. 15: Монтирование каталога NFS на клиенте и проверка через mount

Настройка автоматического монтирования в /etc/fstab

```
GNU nano 8.1 /etc/fstab Modified

#
# /etc/fstab
# Created by anaconda on Sat Sep  6 16:45:04 2025
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
UUID=bf9aaae8-a4dd-4b5a-be57-7faf0fd4a158 / xfs defaults 0 0
UUID=c92e69ed-3ed5-4d9c-ba1a-47af2e3b4400 /boot xfs defaults 0 0
UUID=5072-7B38 /boot/efi vfat umask=0077,shortname=winnt 0 2
UUID=d95619c0-6bd0-463c-8065-1bcb2fcf63ed /home xfs defaults 0 0
UUID=44d16fce-ebea-40da-8e90-ef32b3bf6061 none swap defaults 0 0
#VAGRANT-BEGIN
# The contents below are automatically generated by Vagrant. Do not modify.
vagrant /vagrant vboxsf uid=1000,gid=1000,_netdev 0 0
#VAGRANT-END
server.nsandryushin.net:/srv/nfs /mnt/nfs nfs _netdev 0 0
```

Рис. 16: Настройка автоматического монтирования в /etc/fstab

Проверка статуса remote-fs.target

```
[nsandryushin@client.nsandryushin.net ~]$ systemctl status remote-fs.target
● remote-fs.target - Remote File Systems
   Loaded: loaded (/usr/lib/systemd/system/remote-fs.target; enabled; preset: enabled)
   Active: active since Fri 2025-11-28 19:39:03 UTC; 26min ago
 Invocation: 637b1a84864549f9842695e40331230f
    Docs: man:systemd.special(7)

Nov 28 19:39:03 client.nsandryushin.net systemd[1]: Reached target remote-fs.target - Remote File System>
lines 1-7/7 (END)
```

Рис. 17: Проверка статуса remote-fs.target

Проверка подключенных ресурсов командой mount

```
[nsandryushin@client nsandryushin.net ~]$ mount
/dev/mapper/rl_vbox-root on / type xfs (rw,relatime,seclabel,attr2,inode64,logbufs=8,logbsize=32k,noquota)
devtmpfs on /dev type devtmpfs (rw,nosuid,seclabel,size=4896k,nr_inodes=211266,nodev=755,inode64)
tmpfs on /dev/shm type tmpfs (rw,nosuid,nodev,seclabel,inode64)
devpts on /dev/pts type devpts (rw,nosuid,noexec,relatime,seclabel,gid=5,mode=620,ptmxmode=000)
sysfs on /sys type sysfs (rw,nosuid,nodev,noexec,relatime,seclabel)
securityfs on /sys/kernel/security type securityfs (rw,nosuid,nodev,noexec,relatime)
cgroup2 on /sys/fs/cgroup type cgroup2 (rw,nosuid,nodev,noexec,relatime,seclabel,nsdelegate,memory_recursiveprot)
pstore on /sys/fs/pstore type pstore (rw,nosuid,nodev,noexec,relatime,seclabel)
efivarfs on /sys/firmware/efi/efivars type efivarfs (rw,nosuid,nodev,noexec,relatime)
bpf on /sys/fs/bpf type bpf (rw,nosuid,nodev,noexec,relatime,mode=700)
configfs on /sys/kernel/config type configfs (rw,nosuid,nodev,noexec,relatime)
proc on /proc type proc (rw,nosuid,nodev,noexec,relatime)
tmpfs on /run type tmpfs (rw,nosuid,nodev,seclabel,size=345560k,nr_inodes=819200,nodev=755,inode64)
selinuxfs on /sys/fs/selinux type selinuxfs (rw,nosuid,noexec,relatime)
systemd-1 on /proc/sys/fs/binfmt_misc type autofs (rw,relatime,fd=36,pgrp=1,timeout=0,minproto=5,maxproto=5,direct,pipe_ino=4885)
hugetlbfs on /dev/hugepages type hugetlbfs (rw,nosuid,nodev,relatime,seclabel,pagesize=2M)
queue on /dev/queue type queue (rw,nosuid,nodev,noexec,relatime,seclabel)
debugfs on /sys/kernel/debug type debugfs (rw,nosuid,nodev,noexec,relatime,seclabel)
tracefs on /sys/kernel/tracing type tracefs (rw,nosuid,nodev,noexec,relatime,seclabel)
tmpfs on /run/credentials/systemd-journald.service type tmpfs (ro,nosuid,nodev,noexec,relatime,nosymlinks,seclabel,size=1024k,nr_inodes=1024,mode=700,inode64,noswap)
fusectl on /sys/fs/fuse/connections type fusectl (rw,nosuid,nodev,noexec,relatime)
/dev/sda2 on /boot type xfs (rw,relatime,seclabel,attr2,inode64,logbufs=8,logbsize=32k,noquota)
/dev/sda1 on /boot/efi type vfat (rw,relatime,fmask=0077,dmask=0077,codepage=437,iocharset=ascii,shortnam=0,minnterrors=re mount-ro)
/dev/mapper/rl_vbox-home on /home type xfs (rw,relatime,seclabel,attr2,inode64,logbufs=8,logbsize=32k,noquota)
sunrpc on /var/lib/nfs/rpc_pipefs type rpc_pipefs (rw,relatime)
vagrant on /vagrant type vboxsf (rw,nodev,relatime,iocharset=utf8,uid=1000,gid=1000,.netdev)
server nsandryushin.net:/srv/nfs on /mnt/nfs type nfs4 (rw,relatime,vers=4.2,rsize=262144,wsize=262144,namelen=255,hard,proto=tcp,timeo=600,retrans=2,sec=sys,clientaddr=192.168.1.30,local_lock=none,addr=192.168.1.1,.netdev)
tmpfs on /run/user/1001 type tmpfs (rw,nosuid,nodev,relatime,seclabel,size=172776k,nr_inodes=43194,nodev=700,uid=1001,gid=1001,inode64)
gvfsd-fuse on /run/user/1001/gvfs type fuse.gvfsd-fuse (rw,nosuid,nodev,relatime,user_id=1001,group_id=1001)
portal on /run/user/1001/doc type fuse.portal (rw,nosuid,nodev,relatime,user_id=1001,group_id=1001)
[nsandryushin@client nsandryushin.net ~]$
```

Рис. 18: Проверка подключенных ресурсов командой mount

Bind-монтирование каталога веб-сервера на сервере NFS

```
[nsandryushin@server.nsandryushin.net ~]$ sudo mkdir -p /srv/nfs/www
[sudo] password for nsandryushin:
[nsandryushin@server.nsandryushin.net ~]$ sudo mount -o bind /var/www/ /srv/nfs/www/
[nsandryushin@server.nsandryushin.net ~]$ ls /srv/nfs
www
[nsandryushin@server.nsandryushin.net ~]$
```

Рис. 19: Bind-монтирование каталога веб-сервера на сервере NFS

Проверка отображения нового каталога на клиенте

```
[nsandryushin@client.nsandryushin.net ~]$ ls /mnt/nfs/  
www  
[nsandryushin@client.nsandryushin.net ~]$
```

Рис. 20: Проверка отображения нового каталога на клиенте

Редактирование файла /etc/exports на сервере



```
GNU nano 8.1 /etc/exports
/srv/nfs *(ro)
/srv/nfs/www 192.168.0.0/16(rw)
```

Рис. 21: Редактирование файла /etc/exports на сервере

```
[nsandryushin@server.nsandryushin.net ~]$ sudo exportfs -r
```

Рис. 22: Экспорт каталогов командой exportfs

Проверка наличия каталога www на клиенте

```
[nsandryushin@client.nsandryushin.net ~]$ ls /mnt/nfs/  
www  
[nsandryushin@client.nsandryushin.net ~]$ ls /mnt/nfs/www/  
[nsandryushin@client.nsandryushin.net ~]$
```

Рис. 23: Проверка наличия каталога www на клиенте

Добавление bind-монтирования в /etc/fstab

```
GNU nano 8.1 /etc/fstab

#
# /etc/fstab
# Created by anaconda on Sat Sep  6 16:45:04 2025
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
UUID=bf9aaae8-a4dd-4b5a-be57-7faf0fd4a158 /                xfs     defaults        0 0
UUID=c92e69ed-3ed5-4d9c-ba1a-47af2e3b4400 /boot            xfs     defaults        0 0
UUID=5072-7B38 /boot/efi        vfat    umask=0077,shortname=winnt 0 2
UUID=d95619c0-6bd0-463c-8065-1bcb2fcf63ed /home            xfs     defaults        0 0
UUID=44d16fce-ebea-40da-8e90-ef32b3bf6061 none             swap    defaults        0 0
#VAGRANT-BEGIN
# The contents below are automatically generated by Vagrant. Do not modify.
vagrant /vagrant vboxsf uid=1000,gid=1000,_netdev 0 0
#VAGRANT-END
/var/www /srv/nfs/www none bind 0 0
```

Рис. 24: Добавление bind-монтирования в /etc/fstab

Повторный запуск exportfs на сервере

```
[nsandryushin@server.nsandryushin.net ~]$ sudo exportfs -r  
[nsandryushin@server.nsandryushin.net ~]$
```

Рис. 25: Повторный запуск exportfs на сервере

Проверка доступа к каталогу www на клиенте

```
[nsandryushin@client.nsandryushin.net ~]$ ls /mnt/nfs/  
www  
[nsandryushin@client.nsandryushin.net ~]$ ls /mnt/nfs/www/  
cgi-bin  html  
[nsandryushin@client.nsandryushin.net ~]$
```

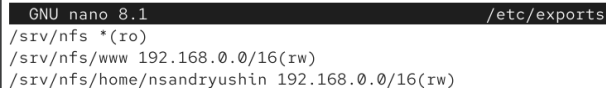
Рис. 26: Проверка доступа к каталогу www на клиенте

Подготовка каталога пользователя и bind-монтирование на сервере

```
[nsandryushin@server.nsandryushin.net ~]$ mkdir -p -m 700 ~/common
[nsandryushin@server.nsandryushin.net ~]$ cd ~/common
[nsandryushin@server.nsandryushin.net common]$ touch nsandryushin@server.txt
[nsandryushin@server.nsandryushin.net common]$ mkdir -p /srv/nfs/home/nsandryushin
mkdir: cannot create directory '/srv/nfs/home': Permission denied
[nsandryushin@server.nsandryushin.net common]$ sudo mkdir -p /srv/nfs/home/nsandryushin
[sudo] password for nsandryushin:
[nsandryushin@server.nsandryushin.net common]$ sudo mount -o bind /home/nsandryushin/common /srv/nfs/home/
nsandryushin
mount: (hint) your fstab has been modified, but systemd still uses
the old version; use 'systemctl daemon-reload' to reload.
[nsandryushin@server.nsandryushin.net common]$ sudo mount -o bind /home/nsandryushin/common /srv/nfs/home/
nsandryushin
[nsandryushin@server.nsandryushin.net common]$ █
```

Рис. 27: Подготовка каталога пользователя и bind-монтирование на сервере

Настройка экспорта каталога пользователя



```
GNU nano 8.1 /etc/exports
/srv/nfs *(ro)
/srv/nfs/www 192.168.0.0/16(rw)
/srv/nfs/home/nsandryushin 192.168.0.0/16(rw)
```

Рис. 28: Настройка экспорта каталога пользователя

Настройка автоматического bind-монтирования в fstab

```
GNU nano 8.1 /etc/fstab

#
# /etc/fstab
# Created by anaconda on Sat Sep  6 16:45:04 2025
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
UUID=bf9aaae8-a4dd-4b5a-be57-7faf0fd4a158 /                xfs     defaults        0 0
UUID=c92e69ed-3ed5-4d9c-ba1a-47af2e3b4400 /boot            xfs     defaults        0 0
UUID=5072-7B38 /boot/efi        vfat    umask=0077,shortname=winnt 0 2
UUID=d95619c0-6bd0-463c-8065-1bcb2fcf63ed /home            xfs     defaults        0 0
UUID=44d16fce-ebaa-40da-8e90-ef32b3bf6061 none             swap    defaults        0 0
#VAGRANT-BEGIN
# The contents below are automatically generated by Vagrant. Do not modify.
vagrant /vagrant vboxsf uid=1000,gid=1000,_netdev 0 0
#VAGRANT-END
/var/www /srv/nfs/www none bind 0 0
/home/nsandryushin/common /srv/nfs/home/nsandryushin none bind 0 0
```

Рис. 29: Настройка автоматического bind-монтирования в fstab

Проверка доступа к каталогу home на клиенте

```
[nsandryushin@client.nsandryushin.net ~]$ ls /mnt/nfs/  
home  www  
[nsandryushin@client.nsandryushin.net ~]$ ls /mnt/nfs/home/  
nsandryushin  
[nsandryushin@client.nsandryushin.net ~]$ █
```

Рис. 30: Проверка доступа к каталогу home на клиенте

Проверка доступа к смонтированному каталогу пользователя на клиенте

```
[nsandryushin@client.nsandryushin.net nsandryushin]$ cd /mnt/nfs/home/nsandryushin/
[nsandryushin@client.nsandryushin.net nsandryushin]$ touch nsandryushin@client.txt
[nsandryushin@client.nsandryushin.net nsandryushin]$ echo "1" >> nsandryushin@client.txt
[nsandryushin@client.nsandryushin.net nsandryushin]$ sudo -i
[sudo] password for nsandryushin:
[root@client.nsandryushin.net ~]# cd /mnt/nfs/home/nsandryushin/
-bash: cd: /mnt/nfs/home/nsandryushin/: Permission denied
```

Рис. 31: Проверка доступа к смонтированному каталогу пользователя на клиенте

Проверка изменений в каталоге common на сервере

```
[nsandryushin@server.nsandryushin.net common]$ ls  
nsandryushin@client.txt  nsandryushin@server.txt  
[nsandryushin@server.nsandryushin.net common]$ cat nsandryushin@client.txt  
1
```

Рис. 32: Проверка изменений в каталоге common на сервере

Подготовка каталогов и создание скрипта nfs.sh на сервере

```
[nsandryushin@server.nsandryushin.net common]$ cd /vagrant/provision/server
[nsandryushin@server.nsandryushin.net server]$ sudo mkdir -p /vagrant/provision/server/nfs/etc
[sudo] password for nsandryushin:
[nsandryushin@server.nsandryushin.net server]$ sudo cp -R /etc/exports /vagrant/provision/server/nfs/etc/
[nsandryushin@server.nsandryushin.net server]$ cd /vagrant/provision/server
[nsandryushin@server.nsandryushin.net server]$ sudo touch nfs.sh
[nsandryushin@server.nsandryushin.net server]$ sudo chmod +x nfs.sh
[nsandryushin@server.nsandryushin.net server]$ sudo nano nfs.sh
```

Рис. 33: Подготовка каталогов и создание скрипта nfs.sh на сервере

Содержимое скрипта автоматической настройки сервера

```
GNU nano 8.1 /vagrant/provision/server/nfs.sh
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install nfs-utils
echo "Copy configuration files"
cp -R /vagrant/provision/server/nfs/etc/* /etc
restorecon -vR /etc
echo "Configure firewall"
firewall-cmd --add-service nfs --permanent
firewall-cmd --add-service mountd --add-service rpc-bind --permanent
firewall-cmd --reload
echo "Tuning SELinux"
mkdir -p /srv/nfs
semanage fcontext -a -t nfs_t "/srv/nfs(/.*)?"
restorecon -vR /srv/nfs
echo "Mounting dirs"
mkdir -p /srv/nfs/www
mount -o bind /var/www /srv/nfs/www
echo "/var/www /srv/nfs/www none bind 0 0" >> /etc/fstab
mkdir -p /srv/nfs/home/nsandryushin
mkdir -p -m 700 /home/nsandryushin/common
chown nsandryushin:nsandryushin /home/nsandryushin/common
mount -o bind /home/nsandryushin/common /srv/nfs/home/nsandryushin
echo "/home/nsandryushin/common /srv/nfs/home/nsandryushin none bind 0 0" >> /etc/fstab
echo "Start nfs service"
systemctl enable nfs-server
systemctl start nfs-server
systemctl restart firewalld
```

Рис. 34: Содержимое скрипта автоматической настройки сервера

Создание скрипта nfs.sh на клиенте

```
[nsandryushin@client.nsandryushin.net nsandryushin]$ cd /vagrant/provision/client
[nsandryushin@client.nsandryushin.net client]$ cd /vagrant/provision/client
[nsandryushin@client.nsandryushin.net client]$ sudo touch nfs.sh
[sudo] password for nsandryushin:
[nsandryushin@client.nsandryushin.net client]$ sudo chmod +x nfs.sh
[nsandryushin@client.nsandryushin.net client]$ sudo nano nfs.sh
```

Рис. 35: Создание скрипта nfs.sh на клиенте

Скрипт настройки NFS на клиенте

```
GNU nano 8.1 /vagrant/provision/client/nfs.sh
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install nfs-utils
echo "Mounting dirs"
mkdir -p /mnt/nfs
mount server.nsandryushin.net:/srv/nfs /mnt/nfs
echo "server.nsandryushin.net:/srv/nfs /mnt/nfs nfs _netdev 0 0" >> /etc/fstab
restorecon -vR /etc
```

Рис. 36: Скрипт настройки NFS на клиенте

Добавление запуска скриптов в Vagrantfile

```
server.vm.provision :script, :scripts => [
  "server nfs",
  type: "shell",
  preserve_order: true,
  path: "provision/server/nfs.sh"
end

## Client configuration
config.vm.define "client", :autostart => false do |client|
  client.vm.box = "rockylinux10"
  client.vm.hostname = "client"
  client.vm.boot_timeout = 1440

  client.ssh.insert_key = false
  client.ssh.username = "vagrant"
  client.ssh.password = "vagrant"

  client.vm.network :private_network,
    type: "dhcp",
    virtualbox____intnet: true

  client.vm.provider :virtualbox do |virtualbox|
    virtualbox.customize ["modifyvm", :id, "--vrdm", "on"]
    virtualbox.customize ["modifyvm", :id, "--vrdmport", "3392"]
  end

  client.vm.provision "client dummy",
    type: "shell",
    preserve_order: true,
    path: "provision/client/01-dummy.sh"

  client.vm.provision "client routing",
    type: "shell",
    preserve_order: true,
    run: "always",
    path: "provision/client/01-routing.sh"

  client.vm.provision "client mail",
    type: "shell",
    preserve_order: true,
    path: "provision/client/mail.sh"

  client.vm.provision "client ntp",
    type: "shell",
    preserve_order: true,
    path: "provision/client/ntp.sh"

  client.vm.provision "client nfs",
    type: "shell",
    preserve_order: true,
    path: "provision/client/nfs.sh"
end
```

Рис. 37: Добавление запуска скриптов в Vagrantfile

В результате выполнения лабораторной работы были получены навыки работы с nfs и сетевыми хранилищами, а так же их настройка